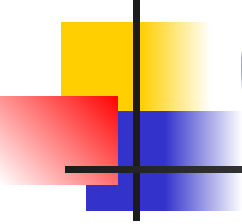


Uniform Circular Motion

Uniform Circular Motion

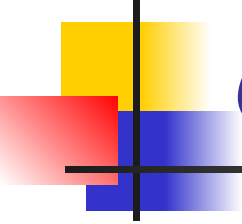
What is Force?

- A force is a **push** or **pull** on an object.
- Forces cause an object to **accelerate**...
 - To speed up
 - To slow down
 - To change direction



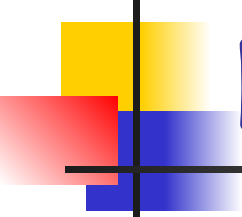
Uniform Circular Motion

- An object that moves at **uniform speed in a circle of constant radius** is said to be in **uniform circular motion**.
- **Question:** Why is uniform circular motion accelerated motion?
- **Answer:** Although the speed is constant, the velocity is not constant since an object in uniform circular motion is continually changing direction.



Centrifugal Force

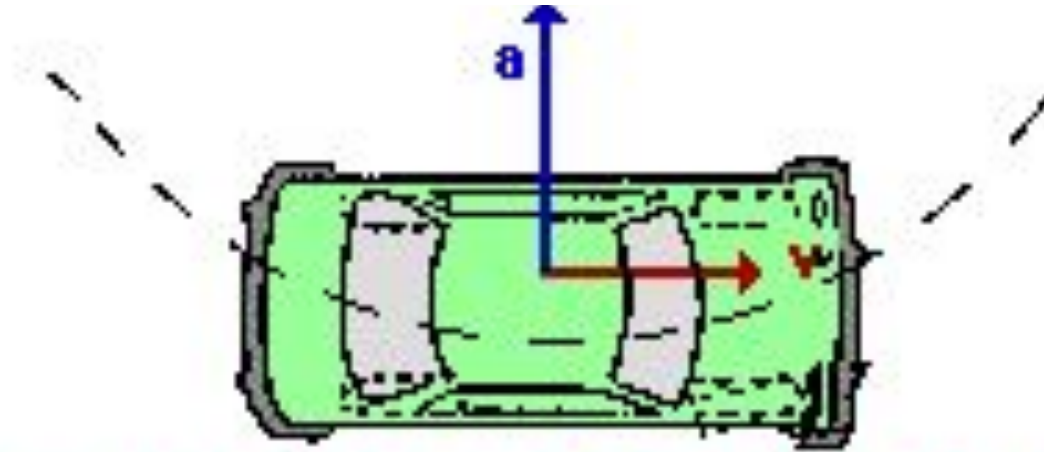
- **Question:** What is centrifugal force?
- **Answer:** That's easy. Centrifugal force is the force that flings an object in circular motion outward. Right?
- **Wrong! Centrifugal force is a myth!**
- There is no outward directed force in circular motion. To explain why this is the case, let's review Newton's 1st Law.



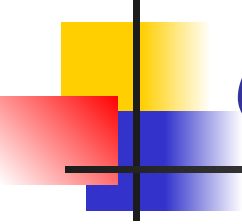
Newton's 1st Law and cars

- When a car accelerates forward suddenly, you as a passenger feel as if you are flung backward.
 - You are in fact NOT flung backward. Your body's inertia resists acceleration and wants to remain at rest as the car accelerates forward.
- When a car brakes suddenly, you as a passenger feel as if you are flung forward.
 - You are NOT flung forward. Your body's inertia resists acceleration and wants to remain at constant velocity as the car decelerates.

When a car turns



- You feel as if you are flung to the outside. You call this apparent, but nonexistent, force "centrifugal force".
- You are NOT flung to the outside. Your inertia resists the inward acceleration and your body simply wants to keep moving in straight line motion!
- As with all other types of acceleration, your body feels as if it is being flung in the opposite direction of the actual acceleration. The force on your body, and the resulting acceleration, actually point inward.



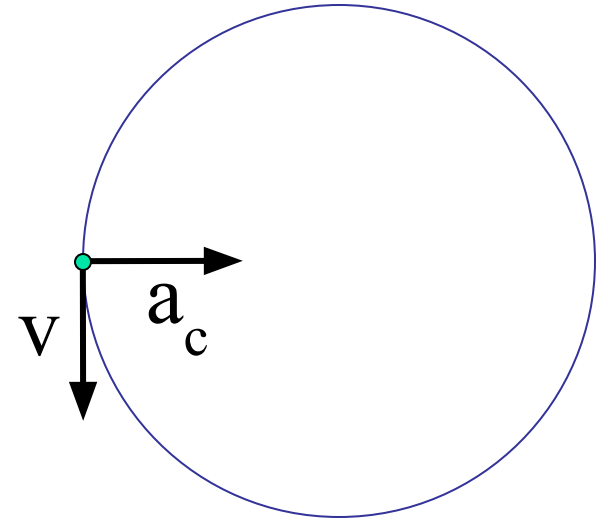
Centripetal Acceleration

- Centripetal (or center-seeking) acceleration points toward the center of the circle and keeps an object moving in circular motion.
- This type of acceleration is at right angles to the velocity.
- This type of acceleration doesn't speed up an object, or slow it down, it just turns the object.

Centripetal Acceleration

$$a_c = \frac{v^2}{r}$$

- a_c : centripetal acceleration in m/s^2
- v : tangential speed in m/s
- r : radius in meters



Centripetal acceleration always points toward center of circle!

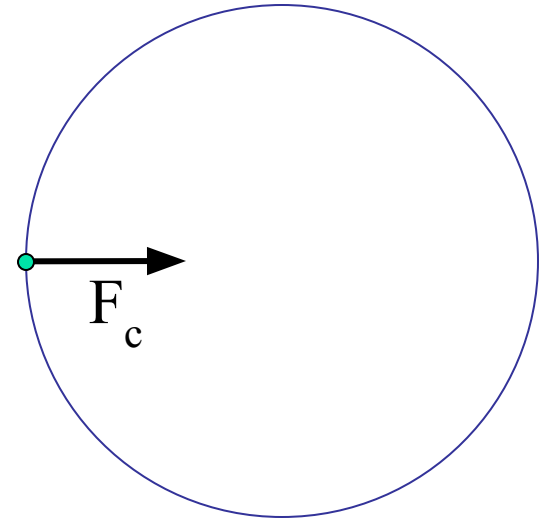
Centripetal Force

- A **net** force responsible for centripetal acceleration is referred to as a centripetal force.
- Centripetal force is simply mass times centripetal acceleration.

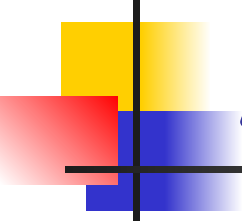
$$F_c = ma_c$$

$$F_c = m \frac{v^2}{r}$$

- F_c : centripetal force in N
- v : tangential speed in m/s
- r : radius in meters

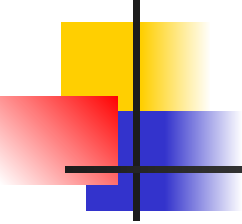


*Always points
toward center of
circle!*



Any force can be centripetal

- The name "centripetal" can be applied to any force in situations when that force is contributing to the object's circular motion.
- You should identify the real force or combination of forces which are causing the centripetal acceleration.
- Any kind of force can act as a centripetal force.

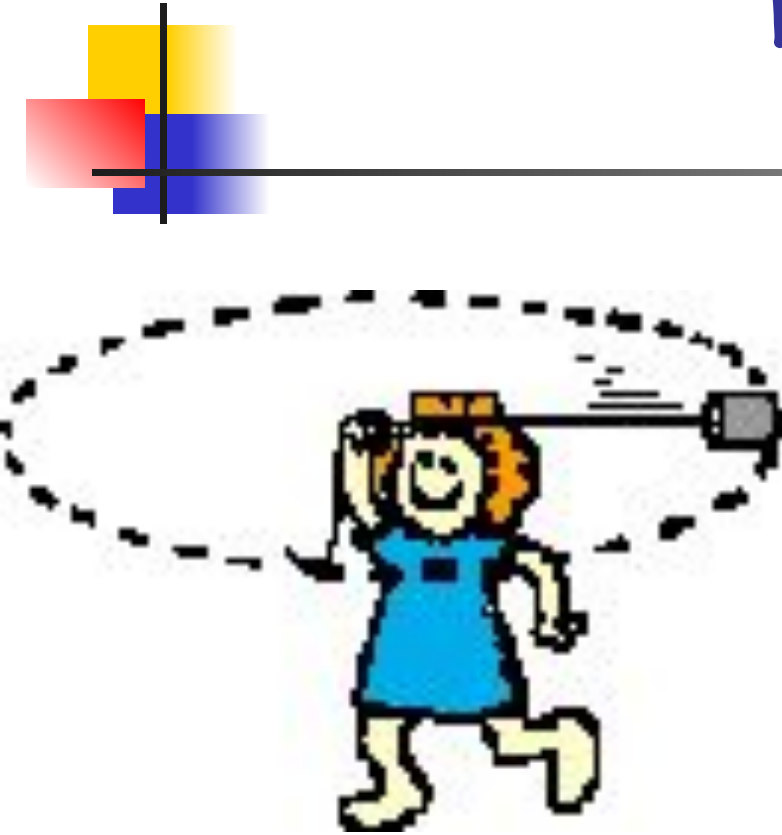


Static friction



As a car makes a turn on a flat road, what is the real identity of the centripetal force?

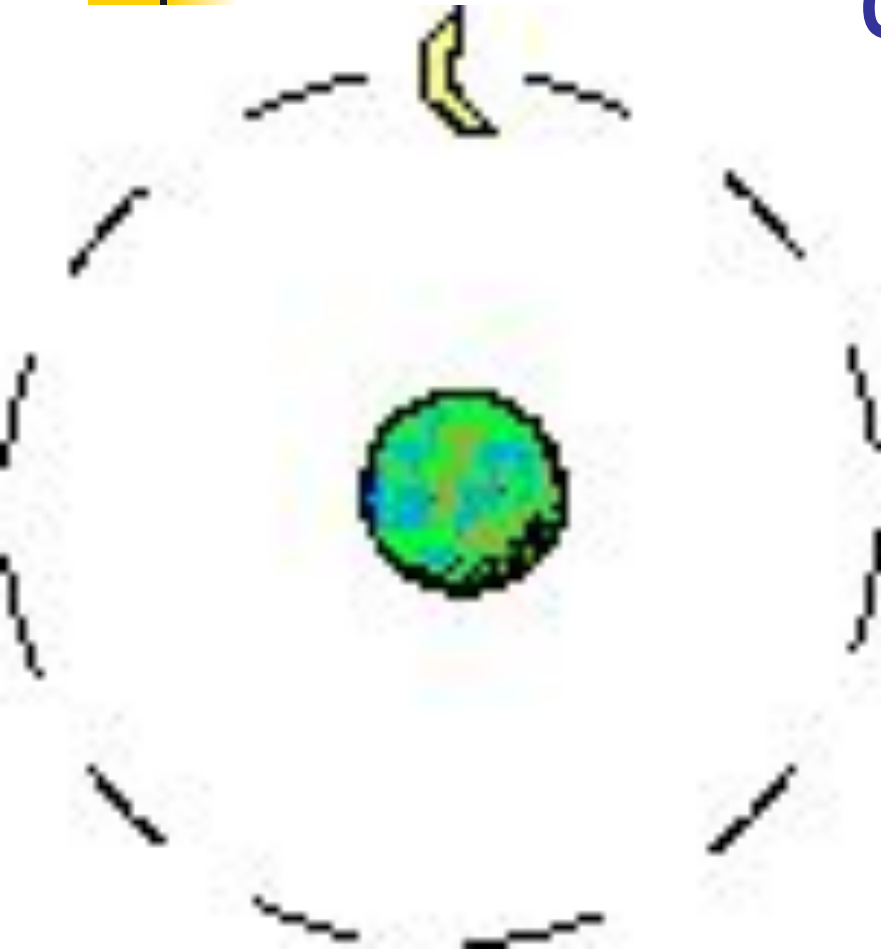
Tension



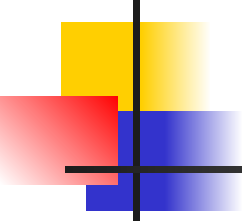
As a weight is tied to a string and spun in a circle, what is the real identity of the centripetal force?



Gravity



As the moon orbits the Earth, what is the real identity of the centripetal force?



Tension, with some help from gravity

As you swing a mace in a vertical circle, what is the true identity of the centripetal force?



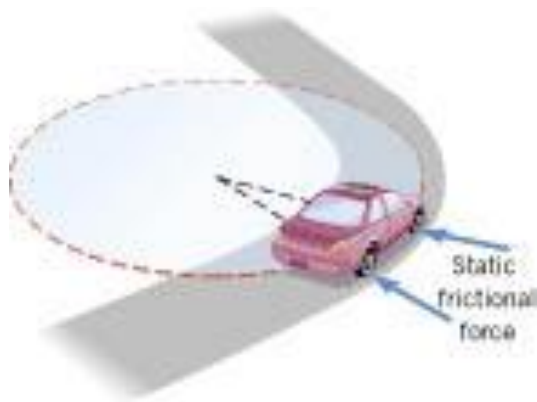
Gravity, with some help from the normal force



When you are riding
the Tennessee
Tornado at
Dollywood, what is
the real identity of
the centripetal
force when you are
on a vertical loop?

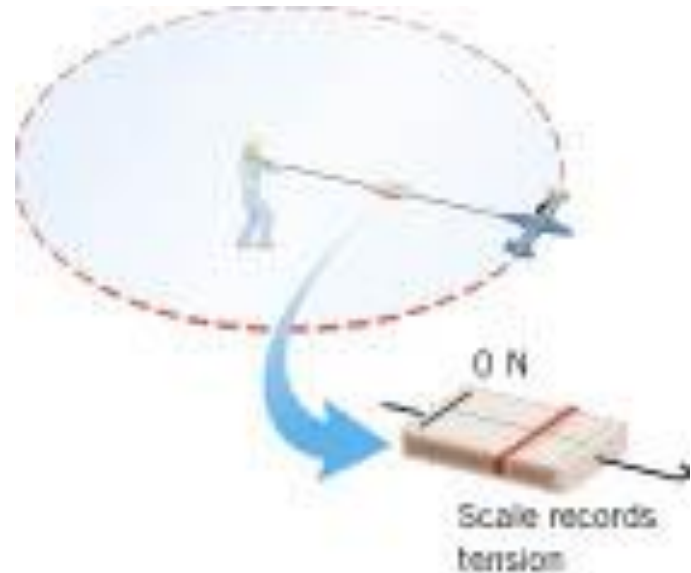
Sample problem

- A 1200-kg car rounds a corner of radius $r = 45$ m. If the coefficient of static friction between tires and the road is 0.93 and the coefficient of kinetic friction between tires and the road is 0.75, what is the maximum velocity the car can have without skidding? When the road is icy, the coefficient of static friction is 0.10. What is the maximum speed the car can take the turn in icy conditions?



Sample Problem

- The model airplane in the figure has a mass of 0.90 kg and moves at a constant speed in a circle that is parallel to the ground. Find the tension T in the guideline for speeds of 19 and 38 m/s . The length of the guideline is 17 m . What is the period for each of the two speeds?



Uniform Circular Motion

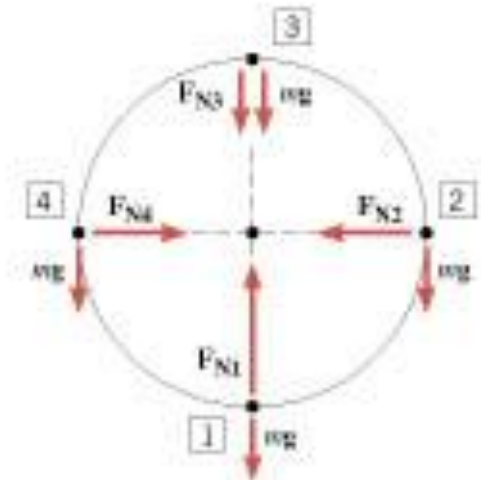
Vertical Circles

Four critical points

- There are four points on a vertical circle where the centripetal force can be easily identified.



(a)



(b)

Sample Problem

- A fighter pilot dives his plane toward the ground at 230 m/s . He pulls out of the dive on a vertical circle. What is the minimum radius of the circle, so that the apparent weight of the pilot never exceeds three times his weight?

Sample Problem

- The condition of apparent weightlessness for the passengers can be created for a brief instant when a plane flies over the top of a vertical circle. At a speed of 215 m/s, what is the radius of the vertical circle that the pilot must use?

Sample Problem

- A 2100 kg demolition ball swings at the end of a 15 m cable on the arc of a vertical circle. At the lowest point of the swing, the ball is moving at a speed of 7.6 m/s. Determine the tension in the cable.